

UC-01 — Start New Game

Use case: Start a new game session and initialize world objects.

Iteration: 1 Last modification: Oct 12 by Amitoj Kochar

Primary actor: Player

Goal in context:

To begin a new game by loading the map, spawning entities, and preparing the player's character with full health and an empty coin count.

Preconditions:

The Game has loaded and is displaying the Main Menu.

Trigger:

The Player selects "Start Game" and a difficulty.

Scenario:

1. The system loads, initializes the map, and applies difficulty settings (sets amount of coins to open exit).
2. The player is spawned at the starting location with full health.
3. Coins, food, zombies, and traps are seeded randomly.
4. Timer starts at 0 and HUD displays health, coin count, and timer.

Exceptions:

1. Invalid difficulty selection → revert to default (Normal).

Priority: Critical (core feature).

Frequency of use: Each new session.

Channel to actor: Keyboard + menu GUI & PC-based program.

Secondary actors: Game engine, RNG spawner.

Channels to secondary actors: Initialization calls and asset loading.

Open issues:

1. Should difficulty also change map size?

UC-02 — Collect Coin

Use case: Player collects a coin item to progress toward escape.

Primary actor: Player

Goal in context:

To gather coins scattered around the map until the required total is reached to unlock the exit door.

Preconditions:

A coin object exists within range.

Trigger: Player collides with a coin object.

Scenario:

1. Collision event detected between player and coin.
2. The System plays coin-pickup sound.
3. The Coin count increments by +1; HUD updates count too.
4. The coin object is removed from the map.
5. The system checks if the target coin count is met. If it is met then it opens the door.

Exceptions:

1. Coin already collected (error) . Thus ignore duplicate event.

Priority: High.

Frequency of use: Continuous throughout the game.

Channels to secondary actors: Collision system and event handlers.

Open issues:

1. Should coins respawn after some time on higher difficulty?
2. Add sound variation for streak collection?

UC-03 — Take Damage (from Zombie or Trap)

Use case: Player takes damage upon contact with enemy or trap.

Goal in context:

To show the loss of health when the player is attacked by a zombie or steps on to a trap.

Preconditions:

Player health > 0 and the damage source is active.

Trigger: Collision between player and damage object.

Scenario:

1. The system detects the collision with either trap or zombie.
2. The system calculates damage value (trap vs zombie).
3. Health reduced by damage; HUD updates.
4. If health > 0, resume normal state.
5. If health $\leq$ 0, invoke UC-06 (Player Death & Restart).

Exceptions:

1. If a zombie collides and a food object also collides (with ever happens first is the case that goes first ex: UC-03 then UC-04)

Open issues:

1. Stack damage if multiple zombies attack in the same frame?

UC-04 — Use Food (Healing)

Use case: Player consumes food to restore health.

Goal in context:

To recover health after taking damage by collecting a food pickup.

Preconditions: Food exists and player health is less than max.

Trigger: Collision with food object.

Scenario:

1. Collision detected between player and food.
2. The System adds fixed heal value to the health.
3. Food object removed from map.
4. HUD updates the health bar.

Exceptions:

1. The Player's health is already full. Then the system ignores pickup.

Open issues:

1. Should food respawn after time on lower difficulty?
2. Add different food types with varying heal amounts?

UC-05 — Reach Exit and Win

Use case: Player escapes once the coin goal is met.

Goal in context:

To complete the objective by reaching the exit door after collecting enough coins (amount is set based on difficulty).

Preconditions:

Player alive (health > 0) and has $\geq$ required coins.

Trigger: Player interacts with exit door.

Scenario:

1. The System verifies the coin requirement is met.
2. Zombies and timer pause.
3. Victory screen shows time, difficulty, coins collected.
4. System saves stats if score system implemented.

Exceptions:

1. Coins < required. Then the message "Collect more coins to unlock exit." is displayed.
2. The Player dies at the door. Then invoke UC-06.

Open issues:

1. If the door is blocked by traps how will the player get in?
2. If the door is blocked by zombies how will the player get in?

UC-06 — Player Death and Restart

Use case: Handle player death and restart mechanic.

Goal in context:

To end a game when health reaches zero and allow restart or menu return.

Preconditions: Health $\leq$ 0.

Trigger: System detects death event.

Scenario:

1. Control input disabled & sound for death plays.
2. Timer and entities stop.
3. Game Over screen shows coins collected and elapsed time.
4. The player chooses Restart, thus returning to UC-01.
5. Or chooses Main Menu, thus returning to the menu screen.

Exceptions:

Frequency of use: Each failed run.

Open issues:

1. Should the timer reset or accumulate over restarts?
2. What if a food object spawns in on the player on the same tick as it is experiencing death?